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Demand Forecasting in the Smart Grid Paradigm: Features and Challenges

Demand forecasting faces challenges that include a large volume of data, increasing number of factors that affect the demand profile, uncertainties in the generation profile of the distributed and renewable generation resources and lack of historical data. A hierarchical demand forecasting framework can incorporate the new technologies, customer behaviors and preferences, and environmental factors.

Mohammad E. Khodayar and Hongyu Wu

I. Introduction

With the advent of deregulation in the power industry, demand forecasting has gained considerable attention in energy market operation and planning. Demand forecasting impacts the operational decisions on generation scheduling, reserve and reliability margins, as well as the maintenance scheduling in electricity networks. Moreover, expansion planning of the generation and transmission assets in power systems is determined based on the provided forecasts on electricity demand (de la Torre et al., 2008). In long-term power system planning, overestimations in demand would result in higher investments in generation and transmission expansions, while underestimations could lead to deficiency in generation supply (Stremel, 1981). Similarly, in

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short-term power system operation, overestimations in demand result in an increased number of committed units to serve the load, which translates into a higher system operation cost, whereas underestimations could result in deficiency in electricity supply and cause catastrophic events as a result of real-time imbalances between generation and demand (Nogales et al., 2002).

Extensive research has addressed the challenges in short-term and long-term demand forecasting in power systems considering the interdependence of the demand patterns with the spatiotemporal behavior of the consumers and weather conditions (Shu et al., 2009; Hong et al., 2013; Nose-Filho et al., 2011; Song et al., 2005; Taylor and Buizza, 2002; Bo and Li, 2009; Fay and Ringwood, 2010). The forecast models developed for utilities over the past two decades were associated with decent economic conditions, which leads to slighter volatility in the load data and parameters affecting customer demand. However, these safe margins were eliminated as the grid approaches its capacity limits, congestion becomes more frequent and public awareness of greenhouse gas (GHG) generation pushes the generation portfolio toward more renewable energy resources. The development of smart grid projects and energy efficiency programs, as well as increased customer awareness on the energy consumption and environmental effects, have dramatic impacts on the models developed for demand forecast in the electricity networks.

In the smart grid paradigm, demand forecasting plays a crucial role, as it applies to several practices, such as demand-side management (DSM), renewable energy integration, and energy storage planning and scheduling. Smart grid deployments, such as energy management systems and advanced metering infrastructure (AMI), provide more efficient real-time control to balance energy supply and demand while offering improved visibility for electricity supply and demand patterns. AMI provides a tool for consumers to understand the correlation between demand and the price of electricity, and offers a better insight into practices in distribution networks such as demand response, pricing scheme, and expansion planning. The widespread development of smart meters provides frequent and continuous readings on demand patterns, which are further aggregated to provide the distribution network demand profile. Several guidelines for investment in generation and distribution infrastructure were developed using information extracted from real-time observations (Park et al., 1999; Firmo and Legey, 2002).

As the number of factors that contribute to demand forecasting increases, the present framework for processing the data requires further revision.

Energy consumption patterns are dependent on time, weather conditions, and random behavior of customers and are further controlled by energy management systems to ensure a certain comfort level for customers. Examples in residential and commercial facilities include heating and refrigeration, space cooling and heating, and air conditioning. In the smart grid paradigm, several technologies contribute to variations in the demand pattern, including renewable energy resources, distributed generation, smart appliances, storage facilities, and electric vehicles (EVs). As the number of factors that contribute to demand forecasting increases, the present framework for processing the data requires further revision. In order to provide more accurate and efficient demand forecasts, several challenges need to be adequately addressed including: (1) processing large volumes of data offered by smart meters corresponding to energy usage; (2) addressing shorter time spans for demand forecasting, i.e., five to 15 minutes; (3) identifying various elements that impact the demand forecasts in the new

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paradigm such as climate change, renewable energy integration, pricing schemes and economic incentives, government policies, and technology developments; and (4) addressing the dynamic behavior of the customers in response to technological development, environmental concerns, and economic incentives.

The needed tools to address the above challenges are required to: (1) process large volumes of historical data in short time spans; (2) develop new frameworks for demand forecasting when the customer behavior is transformed as a result of new practices and technologies in the energy industry, and when the historical data may no longer represent the behavior of consumers; and (3) derive the elements that affect demand and contribute to energy management practices in the smart grid paradigm. The increased number of elements considered in demand forecasting can be captured by developing a set of scenarios reflecting the social, economic, and environmental states of the system, under which the demand forecasting framework is developed. In general, the scenario-based forecasting model requires enterprise-level tools to provide flexibility, scalability, accuracy, and dexterity for both short-term and long-term demand forecasting [SaS White paper]. Examples of such scenarios include severe weather conditions in electricity networks, which affect the availability of the generation and transmission assets as well as demand patterns. The demand pattern in the scenarios addresses how customer demand responds to climate changes in severe weather conditions. Developing such frameworks would help utilities to plan ahead for better reliability and resilience of energy supply in the electricity networks.

The increased number of elements considered in demand forecasting can be captured by developing a set of scenarios reflecting the social, economic, and environmental states of the system.

II. Key Factors in Demand Forecasting

The demand profile in the smart grid deviates from historical patterns as a result of integrating communication, monitoring, and control technologies at the customer level. These technologies include EVs, smart and price-responsive appliances, and renewable energy resources implemented close to the demand side, which change conventional demand patterns and increase the complexity of the demand forecasting methodologies. For example, for price-responsive demand and EV technology, it is expected that most of the charging processes for residential EVs take place at night, which will eventually flatten the demand profile in the distribution network. In addition, integration of distributed renewable energy resources introduces uncertainties to net demand, defined as electricity demand minus renewable generation. As a result of the stochastic nature of renewable energy sources, the uncertainty, volatility, and intermittency of net demand are much higher compared to those of electricity demand, leading to higher ramping requirements for generation assets to follow the net demand in real time. More specifically, the elements that ought to be considered in demand forecasting under the smart grid paradigm are as follows:

1. *Weather*: Accurate weather forecasts lead to more precise renewable generation such as wind and solar, and demand forecasts in the distribution network. The role of the weather is further highlighted once the customers are experiencing extreme conditions, in which a slight change in the weather condition would result in dramatic changes in the comfort level and electricity demand. **Figure 1** shows the correlation between the demand and the temperature in an extreme condition. The demand for electricity in Texas increased to

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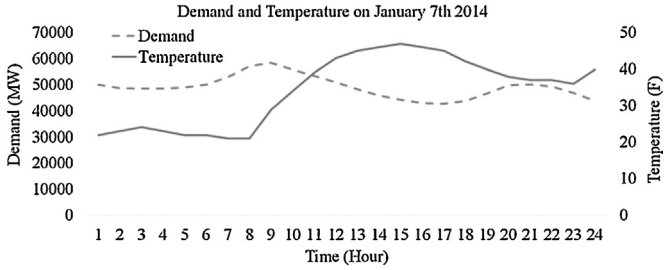


Figure 1: Electricity Demand of the ERCOT on Jan. 7, 2014

57,277 MW at 8:00 a.m. on Jan. 7, 2014, as heaters were turned on when the temperature reached its lowest level. The temperature in the Dallas/Fort Worth area was 21 °F, as shown in the figure. As the temperature increased later in the day, demand decreased once heaters were turned off.

Several factors affect the comfort level of residents including clothing, intensity of activity, individual metabolism rate, and environmental conditions such as ventilation and air velocity, as well as the temperature of the surroundings. Among others, the environmental conditions are controlled by the energy management systems. The outdoor temperature affects the operative indoor temperature; however, a small deviation in indoor temperature or air speed will not affect the thermal comfort of the occupants. Switching on air conditioning and refrigerating systems will not impact the indoor temperature and humidity immediately because of the heat storage capability of the building. Therefore, the electricity demand of air conditioning and refrigeration could be regulated to provide demand response services and the storage capability provides the headroom that affects the demand profile in response to the changes in the weather conditions.

2. *Temporal factors*: Electricity consumption is dependent on time of day, day of the week (weekday, weekend), the type of day (holiday, special event) as well as the season. The demand-side management and pricing mechanism in the smart grid can limit the fluctuation of demand and the difference between peak and off-peak periods. The new demand pattern is much flatter than the traditional demand pattern in the electricity grid, which reduces the temporal impact on the intraday demand forecast profile in the smart grid. However, factors such as seasonal differences, climate changes, weekends and holidays would still impact the value of demand.

3. *Volatility in generation and demand profiles*: Random disturbances in consumption and generation are consequences of customer behaviors and the availability of renewable energy resources, respectively. The fluctuations in weather parameters including cloudiness, wind speed, temperature, and pressure will impact the renewable energy generation portfolio. The integration of new technologies in the smart grid such as EVs and energy storage devices, which have several times larger demand capacity than residential loads, would contribute to the magnitude of the disturbances in the demand pattern. The random disturbances may result in sudden imbalances between generation and demand, which can be compensated by using fast-response generation technologies. Deploying distributed generation (DG) with intelligent control mitigates such fluctuations locally and reduces the required load following and regulation services in the wholesale electricity market.

4. *Electricity price and demand response*: In regulated electricity markets, the electricity price is fixed to recover the cost of energy generation, transmission, and distribution, while in a competitive market structure, price is determined based on the balance between demand and supply. In the latter paradigm, the consumers respond to variations in the price of electricity by adjusting their demand, which is in turn determined by the pricing schemes. These pricing schemes ranging from time-of-use (TOU) to real-time pricing are also referred by terms such as time-differentiated retail pricing, time-variant pricing, and advanced pricing programs. TOU pricing scheme will increase the incentives for the customers to shift peak load to off-peak

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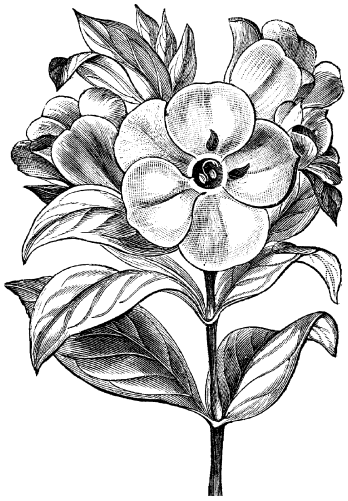
periods. Recent advancements in metering and energy management technologies have increased the appeal of real-time pricing and demand response practices. Such practices could reduce the cost of serving demand in the electricity market, reduce the GHG of generating electricity and change the demand forecast models as they provide incentives to reduce the demand once it exceeds supply. By integrating distributed energy management systems in the distribution networks, these practices can limit the costly generation expansion plan to serve peak demand in short periods. As an example, by implementing demand response in California, demand was reduced by 1,500 MW, i.e., 2 percent of total statewide demand, at peak hours during the summer of 2006. Consequently, no rolling blackouts were ordered during peak demand periods that summer. Analogously, ISO New England, Exelon Corp., and Northeast Utilities reduced their respective demand by 530 MW, 1,100 MW, and 95 MW by implementing demand response practices. The total 37,500 MW potential resources contributed by demand response would have dramatic impacts on demand forecasts (Federal Energy Regulatory Commission, 2006). Integrating demand response will change demand forecast schemes as the correlation between demand and electricity price is further highlighted and available resources will impact the extent of these services. The availability of these resources is dependent on such developing technologies on the demand side as smart appliances and energy storage facilities behind the meter.

5. *Renewable energy resources*: The smart grid promotes the integration of renewable energy resources, primarily wind and solar generation. Wind and solar energy are reliable sources for electricity, are deployable at small scale, and can diversify our nation's energy portfolio. The available generation resources close to the demand side will affect the net demand portfolio that is served by utilities. Any deficiency in supply as a result of volatility in renewable resources can be compensated by electricity purchased at a higher price which is provided by more expensive dispatchable generation assets in the real-time electricity market. Hence, developing renewable energy resources close to the demand side imposes further imbalances between generation and demand. Such technologies increase the net demand uncertainties and increase net demand forecast errors; however, the level of uncertainty is dependent on the generation profile and its correlation with the demand profile. For example, some aspects of solar energy fluctuation including sunrise and sunset are predictable, the generation profile is close to the demand profile as the peak occurs during the day and the intermittency is affected by the cloud coverage in the area. Unlike solar generation, the wind generation pattern is more volatile and intermittent and may not coincide with the demand pattern. However, there is a strong complementarity between these two resources even though the level of such complementarity depends on location (Li et al., 2009). Moreover, aggregation of such resources would result in the reduction in the intermittency and volatility of the overall generation profile. Other renewable generation technologies including hydro and biomass could provide more stable generation profiles. Generation patterns of renewable resources including small wind generation and rooftop solar panels rely on such weather conditions as solar insolation, cloudiness of the sky, temperature, humidity, and wind speed profile. More accurate weather forecasts improve the generation forecast of such resources.

There is a strong complementarity between wind and solar resources even though the level of such complementarity depends on location.

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6. *Energy storage*: Advancements in energy storage technologies enables customers to have large-scale, efficient storage facilities for electricity. Energy storage provides flexibility in both operations and planning in the presence of the variable renewable generation in the grid. The generation pattern for renewable energy resources is not in direct correlation with the energy consumption. Usually, wind speed is higher at night, when demand is low. Thus, the wind generation is stored in storage units and retained for use in peak periods. The energy storage facilities could help shift the demand profile from peak periods to off-peak periods. The storage unit is charged during off-peak periods, when demand and electricity price are low, and discharged during peak periods once demand and the price of electricity are high. The capacity of the storage facilities is dependent on the scale of demand and available renewable resources. Utility demand is affected by the amount of storage capacity developed. The energy storage facilities would change the forecasted demand profile, as the operation scheduling of these units relies on the electricity price and their technical constraints, including the limitations on capacity and ramping capabilities. Furthermore, energy storage can facilitate demand response practices in distribution networks. Behind the meter, storage facilities not only improve the reliability of energy supply but also increase the demand response capability of distribution networks. Such storage facilities become more popular as a cost-effective solution to reduce the peak demand and charges among commercial and industrial businesses. This market is growing rapidly, as over 850 MW of distributed energy storage will be annually deployed



in the U.S. by 2020 (US Energy Storage Market, 2015). One of the major challenges corresponding to such storage facilities is the limited information on their operation schedule. Such units are mostly owned and operated by customers to reduce their peak demand charges as well as the cost of energy, and their operation schedule is not readily apparent to utilities. Hence, such storage facilities may impose large errors in the demand forecast of customers unless they are monitored exclusively.

7. *Electric vehicles*: EVs are considered a favorite alternative for transportation with minimum dependence on the oil and gas market. As the penetration of EVs in the market increases, the demand profile eventually changes as a result of charging and discharging patterns. Charging EVs will increase system load while integrating the vehicle to grid (V2G) technology can regulate demand in distribution networks and increase the reliability of energy supply. Nonetheless, there are several regulatory, technical, implementation, and market barriers for developing this technology. Furthermore, the stochastic nature of customer behavior, including randomness in arrival time, charging time, and the expected charging demand, would increase the uncertainty in the forecasted demand profile. The state of the distribution network can also contribute to the forecasted demand of EVs. For example, after prolonged outages in a network with high penetration of EVs as well as the heat and electricity storage devices, there is a sudden increase in the demand profile as a result of charging EVs and energy storage facilities. These sudden peaks should be addressed in the forecasted demand in severe conditions after restoration of the network. If the system is in normal operation, this peak can be avoided by scheduling the charging cycles, using the V2G capability of the EVs and regulating demand using energy management systems. Some studies have indicated that large number of EVs can be

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accommodated without increasing the peak demand of the distribution network by coordinating the charging pattern of the EVs. The peak demand can further decrease by integrating V2G technology in the system (Rassaei et al., 2014). The price signals from the utility provide incentives for the EVs to charge during off-peak hours, which increases total demand, and to discharge (if required) at peak hours, which reduces demand and price. A study in PG&E showed that the TOU program influenced the charging pattern of EVs and the customers changed the utility rate subscription as a result of obtaining EVs (EV, nd). Moreover, the charging/discharging cycles are coordinated with demand response practices in distribution networks to reduce the peak demand and decrease the loading of the transformers in the distribution network, while maintaining an acceptable comfort level for the customers (Shao et al., 2012).

8. *Smart and efficient appliances*: The residential and commercial sectors consumed 70 percent of total electricity generation in 2013 (Annual Energy Outlook, 2015). The developed energy management technologies in these sectors bring intelligence to energy consumption. In residential level, home energy management systems (HEMS) communicate with smart home appliances and smart electric meters to help curtail and shift electricity demand according to the electricity price signals. Conventional appliances, such as air conditioners, refrigerators, and dishwashers, could be converted into smart appliances by adding monitoring and control modules. Shifting demand to off-peak hours would yield a reduction in energy costs with TOU or real-time electricity



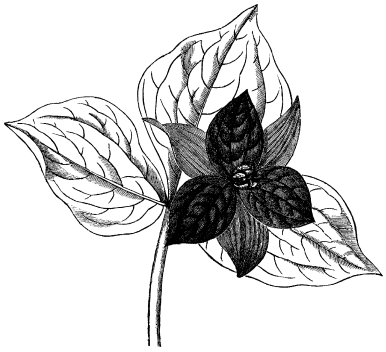
pricing schemes. Over the past years, smart appliances with sophisticated control and monitoring interfaces that could be installed on cell phones and computers were introduced by several companies; yet the smartness in energy consumption is only associated with lowering home energy bills. The crucial need for introducing smart appliances is further highlighted in controlling large appliances. Electricity consumption by heating, ventilation, and air conditioning (HVAC) and refrigeration systems accounts for a large portion (50–60 percent) of the annual energy consumption of commercial buildings. Hence, the energy management of such facilities can lead to significant savings for the utility that is further shared with customers. In smart grids, the utilities are required to understand the behavior of the customers in operating their air conditioners, pool pumps, TV sets, dish washers, washers and driers, water heaters, EVs and other appliances and utilize the gained knowledge to provide more sophisticated forecasting methodology, considering the abundance of data inputs from a data-rich smart grid environment. Advances in the technologies result in a decrease in demand growth as energy efficiency measures are improved among customers. Increasing the number of STAR appliances or more efficient LED lighting facilities contributes to reduction in the growth of energy consumption, which in turn reduces peak demand. This would also affect the demand forecast measures in the electricity networks. The objective of demand management in the industrial sector is to maximize the system load factor, and reduce power losses and peak demand. Similar energy management techniques are used in this sector with corresponding requirements of the facilities.

9. *Virtual power plants, microgrids and nanogrids*: DG provides new opportunities to serve the demand with low-capacity electricity generation technologies placed close to end customers. As local generation

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serves demand behind the meter, the demand forecast schemes have to incorporate such controlled generation resources. If the DGs are renewable energy resources, the stochasticity in the generation profile will make the forecasting framework more complicated, as the outcomes of the net demand forecast are dependent on the forecasted renewable energy profile. In smart grids, the virtual power plant (VPP) is referred to as a group of customers under a specific demand response or pricing program who respond to the energy market as a unified flexible source. Incorporating VPPs allows for treating the demand entities as another operational plant with its respective attributes, including maximum capacity, minimum capacity, ramp-up and ramp-down capability. The demand portfolios of VPPs are regulated by leveraging the local dispatchable generation assets. Increasing energy efficiency through development of combined heat and power (CHP) technologies provides improved flexibility of energy supply for demand with different characteristics (i.e., electricity and heat) in the distribution networks.

Microgrids are defined as aggregated demand and generation sources that operate as a single system with specified boundaries. The monitoring and control assets in the microgrid facilitate autonomous responsiveness to disturbances in the power grid. While microgrids and VPP share critical characteristics, they are different in several aspects. Microgrids can operate autonomously by islanding in case there is an incident in the utility grid, while VPPs are always grid-connected. Moreover, microgrids rely on storage facilities, and are connected to distribution networks, while VPPs



may not have any storage and can participate in the wholesale electricity market. Hence, microgrids can also form VPPs while VPPs are not necessarily microgrids (Asmus, 2010). Integrating microgrids provides the required flexibility, accessibility, and economic and reliability benefits introduced by the smart grid and promotes the distributed intelligence to operate the electricity network efficiently. Providing the monitoring and control requires real-time information on generation and demand patterns, which further calls for online adaptation of generation and demand forecast frameworks.

Nanogrids are small microgrids that serve a single load and support a wide range of applications from emergency to basic power supply. Components of a nanogrid are a controller, loads, storage (if any), and gateways. Loads in a nanogrid can be any electrical device, commonly in the range of 1–100 W in power demand (Nordman and Christensen, 2013). Electricity usage varies in response to the function or state of the devices. The level of power supplied to its loads is controlled by the nanogrid controller, which is able to negotiate with other nanogrids and microgrids through gateways, set the local electricity price, and manage internal storage. The load can be fully served, partially served, or denied by the nanogrid controller. The controller uses the past energy usage pattern to determine the load preferences and sets the decision variables to correspond to the supply assets. If an energy storage facility is attached to a nanogrid, the controller stores or withdraws electric energy as needed within the storage capabilities (Nordman and Christensen, 2013). Nanogrids are considered to be the building blocks of the more complex microgrids or utility-dominated smart grids. Since nanogrids serve a small number of loads, they leverage fast local monitoring and control for energy management. The provided control at this level could benefit the developed

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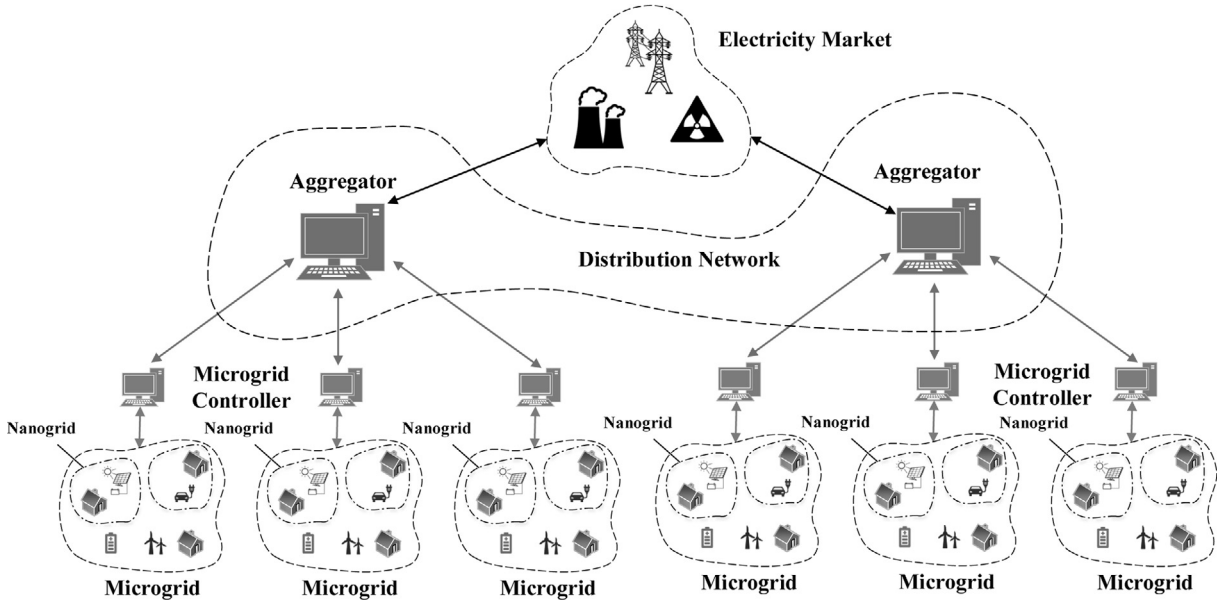


Figure 2: The Hierarchical Structure of Nanogrids and Microgrids in Power Systems [designed by Freepik]

demand forecast framework as demand is being controlled at very fundamental levels with available generation resources. At the higher level, the microgrid controllers will perform energy management to regulate demand in the distribution network. Accordingly, the forecasting approach for generation and demand is affected by the developed monitoring and control at each level. Such controls will shape demand and impact the forecasting schemes accordingly. Figure 2 shows the hierarchical structure of the nanogrid and microgrid in distribution networks. At a higher level, the forecasted demand is dependent on historical data of net demand, which is further regulated at lower levels. Developing the energy management framework at each level will impact the demand forecasting respectively.

III. Challenges and Solutions

The current forecast methods in the power network are based on conventional generation technologies and passive customers. However, in the smart grid paradigm, the framework to provide demand forecast should change significantly to address the customers' awareness on energy consumption and their participation. The forecast of energy consumption patterns is performed at three levels: (1) very short-term load forecasting (VSTLF), which has short time spans from seconds to hours; (2) short-term load forecasting (STLF), which has a longer time horizon from hours to weeks, and (3) long-term load forecast (LTLF), which has a time horizon from months to years. Hence, the developed demand forecast models should improve forecasting efficiency and accuracy at all levels.

There are several challenges associated with forecasting model development in the smart grid. A major one is associated with the large volume of available data for electricity consumption in small time periods provided by the smart meters (i.e., five to 15 minutes), which requires efficient data processing algorithms and effective computing assets to process such large volumes in a reasonable amount of time. Another challenge corresponds to the number of elements that affect electricity demand at the distribution level. Such factors include environmental, economic, and social factors that increase the complexity of the forecast model and may lead to over-fitting the model. The third challenge is the intricacy in determining which factors contribute to the changes in electricity demand at each time period. An increase in off-peak demand might be a result of factors such as charging EVs and

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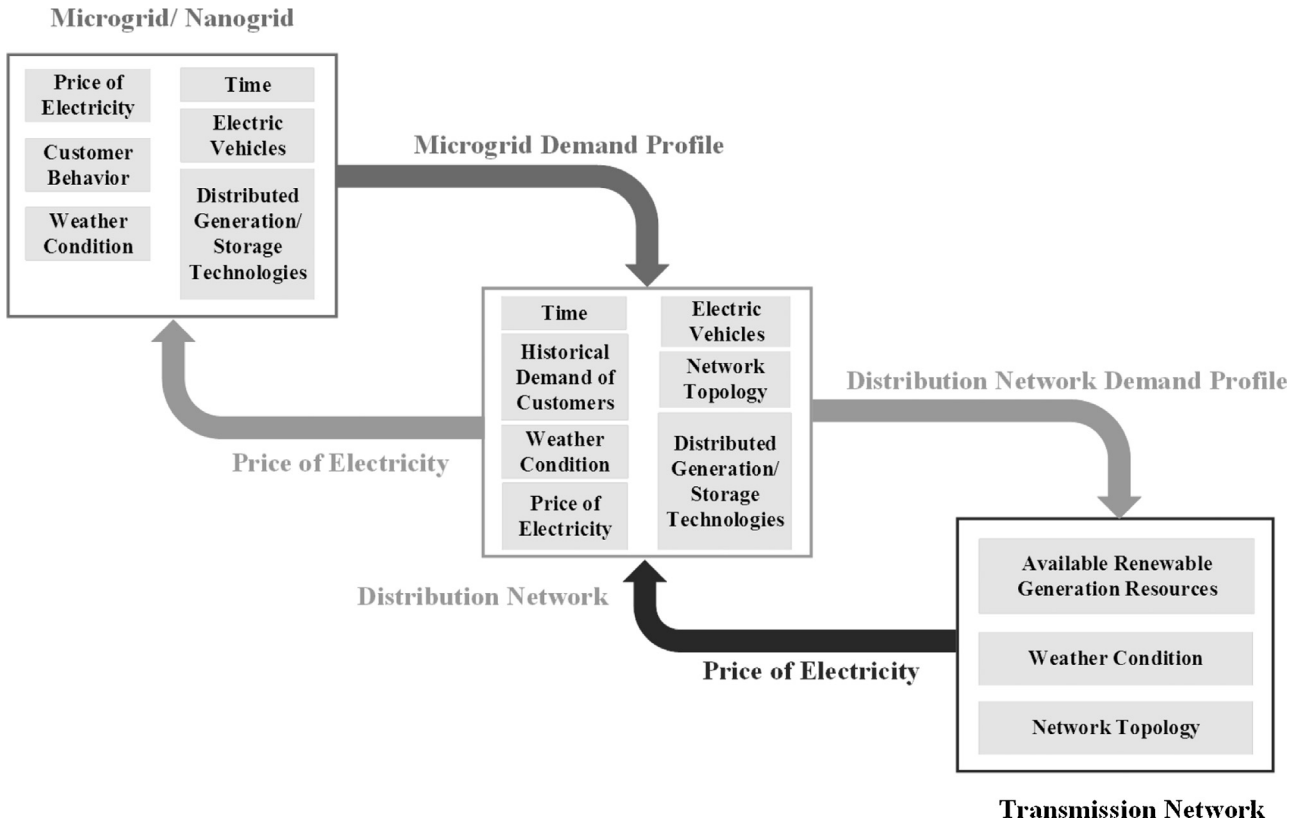


Figure 3: The Hierarchical Structure of Developing the Demand Forecast in the Smart Grid Paradigm

storage facilities behind the meters deployed by customers. Similarly, a reduction in peak demand might be caused by discharging storage facilities, EVs, or the exercise of demand response. The fourth challenge is forecast errors in measurements, including temperature, humidity, renewable generation, and the customer participation level in demand response. The accuracy of the forecasted data is decreased as more uncertain factors contribute to forecasted values in the smart grid paradigm. Last but not least, the deficiency in the volume of the data to fathom the impact of different elements and behaviors on electricity consumption pattern is a big challenge to ensure the accuracy of the developed forecast model. For example, in order to realize the customer behavior with respect to the variation in the price of electricity, the required historical data to model demand and pricing schemes may not be commonly available; such deficiencies would be reflected in the training sets used to develop the model.

In short, the emerging smart grid environment requires distributed intelligence and new models based on artificial intelligence and neural networks (ANNs), expert systems, fuzzy logic, and support vector machines, for accommodating the new contributors that affect the demand forecast. The required forecasting models address the climatic, social, economic, and habitual factors that have direct impact on the demand and generation profile in a distributed environment. The effect of such factors is less realized in the aggregated demand profiles. For example, the load profile of an independent system operator (ISO) has a much smoother profile than that of distributed environments (distribution network feeder, microgrid, nanogrid, etc.). The demand profile in the distributed environment is more volatile; however, it reflects more information on the correlation between environmental factors and electricity demand. Hence, differentiating the effect of environmental factors on electricity demand is more difficult on the aggregated demand profile as it covers diverse geographic areas with

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various environmental conditions.

Figure 3 shows the hierarchical structure of the demand forecast in the smart grid paradigm. It shows how the demand forecast at each level leverages the procured forecast at its lower level. For instance, the demand forecast at the microgrid/nanogrid level procures the demand served by local generation resources; and the demand profile of the nanogrid/microgrids realized from the utility side reflects the intelligence to perform the energy management at this level. At the higher level, the developed forecasting methods for the distribution network takes into account the historical data of the customers that are not clustered in microgrids, the forecasted demand of the nanogrids/microgrids, the available distributed generation resources including renewable technologies, as well as the topology of the network, all of which correspond to the available generation and demand resources in the distribution network. In the wholesale electricity market, the net demand profile is determined by aggregating the demand profile of the distribution networks and the forecasted renewable energy resources. At this level the demand forecast performed by distribution network operators can be used by demand aggregators to procure the demand profile of the wholesale electricity market, and the forecasted renewable energy resources will contribute to the net demand profile of the system. The main feedback from the wholesale electricity market to the distribution networks and the nanogrid/microgrid level is the price of electricity, which impacts the forecasted demand profile at those levels.

IV. Transition to the Smart Grid Era

In order to achieve an acceptable forecast accuracy, demand forecasters try to understand the behavior of customers in response to economic and environmental factors. In the transition toward the smart grid via the implementation of smart meters, there exist customers with traditional meters in the network. Since the behaviors of these customers are different from those equipped with smart meters, the demand profile of overall customers is a combination of the demand profiles of both types of customers. Therefore, any forecast model should be able to differentiate the type of customers with their respective characteristics. The time span for recording and processing the demand for smart meters is as small as 15 minutes. Thus, the forecast model can be developed to forecast demand for the next 15, 30, or 45 minutes; yet for the traditional meters, it is not possible to develop the forecast model for shorter time spans as the data is not available for such periods. Consequently, utility forecasters need to analyze the time span for the forecast model in view of the mixed consumption pattern of the two categories of customers. As the amount of available data in the smart grid is quite large, the forecasting model should be also able to process large volumes of data by providing accurate data filtering and by pre-processing and using a combination of statistical and data driven techniques to increase the accuracy of the demand forecasts. Accurate load forecasting using the large amount of historical data is very critical in building the hierarchical model for demand forecasting in the smart grid paradigm. The increase in the amount of sampling data provided by smart meters would increase the efficiency and accuracy of the short-term load forecasting. Moreover, an increase in the number of sampling points would improve the forecast model as the data correspond to the assets that contribute to the demand forecast were collected exclusively. For example, measuring the dispatch of energy storage facilities installed behind the meters in response to the price of electricity, and collecting data on demand response capability and behavior of the customers in response to the price of electricity independently, would highlight the role of each asset and practice

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in shaping the demand profile and improve the developed demand forecast models accordingly.

II. Conclusion

More advanced demand forecast techniques would result in improvement in the operation and maintenance of electricity networks, and provide safety, security, sustainability, and energy efficiency to the electricity infrastructure. Introducing local variables that have a direct impact on the demand and generation profile will increase the complexity of the forecast models in the smart grid. Several factors that contribute to the demand forecasting in the smart grid paradigm include distributed energy resources installed at the customer side, storage facilities behind the meters, and hierarchical measurement and controls in the distribution networks, for which the forecasting model at the aggregated levels relies on the outcome of the forecast demand on the lower levels. This hierarchical structure of demand forecast can efficiently addresses the effect of local variables on the aggregated demand profile in the smart grid. The electricity demand is dependent on the sector (industrial, commercial, and residential), and the response of demand at each sector to such environmental and economic factors is different; therefore, the forecasting model developed for each sector ought to incorporate the corresponding characteristics. Such a hierarchical structure could be extended to address short- and long-term demand forecasts in the smart grid.■

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